

Training Loop in Deep Learning

Backpropagation

Chain Rule:

$$\frac{\partial}{\partial x} g(f(x)) = \frac{dg}{df} \frac{df}{dx}$$

* forward propagation *

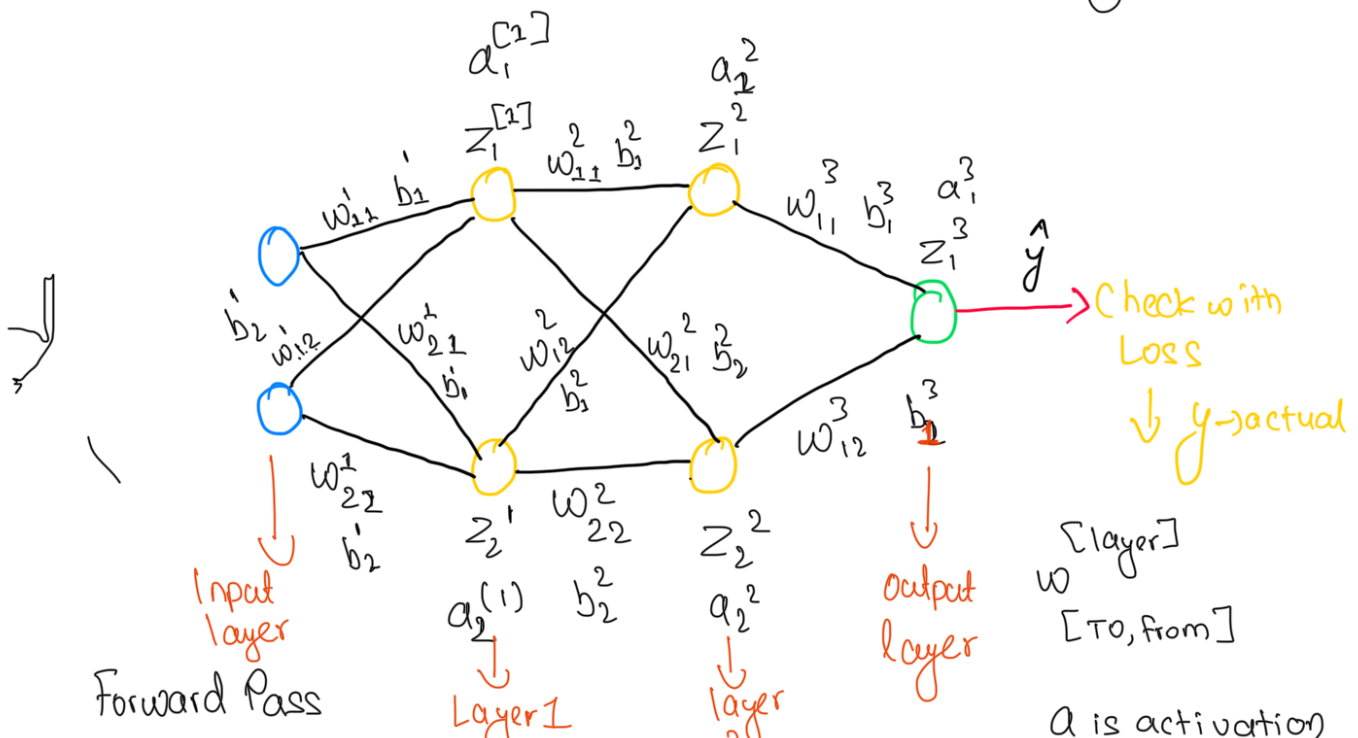
Architecture

Input layer: x_1, x_2

Layer 1 (hidden layer) = 2 Neuron ($n_1^{[1]}, n_2^{[1]}$)

layer 2 (hidden layer) = 2 Neuron ($n_1^{[2]}, n_2^{[2]}$)

Output = 1 Neuron ($\hat{y}$)



function

① Hidden layer 1

$$z_1^1 = w_{11}^1 x_1 + w_{12}^1 x_2 + b_1^1$$

$$a_1^1 = \text{ReLU}(z_1^1) \rightarrow \text{act as input for other hidden layer}$$

$$z_2^1 = w_{21}^1 x_1 + w_{22}^1 x_2 + b_2^1$$

$$a_2^1 = \text{ReLU}(z_2^1) \rightarrow \text{act as input for other hidden layer}$$

② Hidden Layer 2

$$z_1^2 = w_{11}^2 a_1^1 + w_{12}^2 a_2^1 + b_1^2$$

$$a_1^2 = \text{ReLU}(z_1^2)$$

$$z_2^2 = w_{21}^2 a_1^1 + w_{22}^2 a_2^1 + b_2^2$$

$$a_2^2 = \text{ReLU}(z_2^2)$$

③ Output layer

$$z_1^3 = w_{11}^3 a_1^2 + w_{12}^3 a_2^2 + b_1^3$$

$$a_1^3 = \text{Sigmoid}(z_1^3)$$

Backward pass

* Backpropagation *

Cost function:

Binary cross Entropy (L)

Step 1: Output Layer [layer 3]

The loss function

$$\delta_1^{[3]} = \frac{\partial L}{\partial z_1^{[3]}}$$

$$\text{where, } \delta_1^{[3]} = \hat{y} - y$$

Weight Gradient

$$\text{Weight 1 } (w_{11}^{[3]}) : \delta_1^{[3]} \cdot a_1^{[2]}$$

$$\text{Weight 2 } (w_{12}^{[3]}) = \delta_1^{[3]} \cdot a_2^{[2]}$$

$$\text{bias } (b_1^{[3]}) = \delta_1^{[3]}$$

Step 2: Hidden Layer 2

(i) Gradient at Layer 2

Error at Node 1 :

$$\delta_1^{[2]} = (\delta_1^{[3]} * w_{11}^{[3]}) * f'(z_1^{[2]})$$

Error at Node 2 :

$$\delta_2^{[2]} = (\delta_1^{[3]} * w_{12}^{[3]}) * f'(z_2^{[2]})$$

(ii) Gradient for weight

$$\text{Node 1: } \frac{\partial L}{\partial w_{11}^{[2]}} = \delta_1^{[2]} \cdot a_1^{[1]}, \quad \frac{\partial L}{\partial w_{12}^{[2]}} = \delta_1^{[2]} \cdot a_2^{[1]}$$

Node 2:

$$\frac{\partial L}{\partial \omega_{21}^{[2]}} = \delta_2^{[2]} \cdot a_1^{[1]}, \quad \frac{\partial L}{\partial \omega_{22}^{[2]}} = \delta_2^{[2]} \cdot a_2^{[1]}$$

Biases:

$$\frac{\partial L}{\partial b_1^{[2]}} = \delta_1^{[2]}, \quad \frac{\partial L}{\partial b_2^{[2]}} = \delta_2^{[2]}$$

Step 3: Gradient at Hidden layer 1

i) Error at Node 1:

$$\delta_1^{[1]} = \left[\left(\delta_1^{[2]} \cdot \omega_{11}^{[2]} + \delta_2^{[2]} \cdot \omega_{21}^{[2]} \right) \right] \cdot f'(z_1^{[1]})$$

ii) Error at Node 2

$$\delta_2^{[1]} = \left[\left(\delta_1^{[2]} \cdot \omega_{12}^{[2]} + \delta_2^{[2]} \cdot \omega_{22}^{[2]} \right) \right] \cdot f'(z_2^{[1]})$$

(Gradient for layer 1 weights)

weight (Node 1):

$$\frac{\partial L}{\partial \omega_{11}^{[1]}} = \delta_1^{[1]} \cdot x_1, \quad \frac{\partial L}{\partial \omega_{12}^{[1]}} = \delta_1^{[1]} \cdot x_2$$

weight (Node 2):

$$\frac{\partial L}{\partial \omega_{21}^{[1]}} = \delta_2^{[1]} \cdot x_1, \quad \frac{\partial L}{\partial \omega_{22}^{[1]}} = \delta_2^{[1]} \cdot x_2$$

Biases

$$\frac{\partial L}{\partial b_1^{[1]}} = \delta_1^{[1]}, \quad \frac{\partial L}{\partial b_2^{[1]}} = \delta_2^{[1]}$$

$$\partial b_1^{[1]}$$

$$\partial b_2^{[1]}$$

Phase 3: Update weight

$$\text{Rule} = w_{\text{new}} = w_{\text{old}} - \eta \cdot \text{Gradient}$$

Layer 3:

$$w_{11}^{[3]} \leftarrow w_{11}^{[3]} - \eta \frac{\partial L}{\partial w_{11}^{[3]}}$$

$$w_{12}^{[3]} \leftarrow w_{12}^{[3]} - \eta \frac{\partial L}{\partial w_{12}^{[3]}}$$

$$b_1^{[3]} \leftarrow b_1^{[3]} - \eta \frac{\partial L}{\partial b_1^{[3]}}$$

Layer 2:

$$w_{11}^{[2]} \leftarrow w_{11}^{[2]} - \eta \frac{\partial L}{\partial w_{11}^{[2]}}, w_{12}^{[2]} \leftarrow w_{12}^{[2]} - \eta \frac{\partial L}{\partial w_{12}^{[2]}}$$

$$w_{21}^{[2]} \leftarrow w_{21}^{[2]} - \eta \frac{\partial L}{\partial w_{21}^{[2]}}, w_{22}^{[2]} \leftarrow w_{22}^{[2]} - \eta \frac{\partial L}{\partial w_{22}^{[2]}}$$

$$b_1^{[2]} \leftarrow b_1^{[2]} - \eta \frac{\partial L}{\partial b_1^{[2]}}, b_2^{[2]} \leftarrow b_2^{[2]} - \eta \frac{\partial L}{\partial b_2^{[2]}}$$

Layer 1:

$$w_{11}^{[1]} \leftarrow w_{11}^{[1]} - \eta \frac{\partial L}{\partial w_{11}^{[1]}}, w_{12}^{[1]} \leftarrow w_{12}^{[1]} - \eta \frac{\partial L}{\partial w_{12}^{[1]}}$$

$$w_{21}^{[1]} \leftarrow w_{21}^{[1]} - \eta \frac{\partial L}{\partial w_{21}^{[1]}}, w_{22}^{[1]} \leftarrow w_{22}^{[1]} - \eta \frac{\partial L}{\partial w_{22}^{[1]}}$$

$$b_1^{[1]} \leftarrow b_1^{[0]} - \eta \sigma_1^{[1]}, \quad b_2^{[1]} \leftarrow b_2^{[0]} - \eta \sigma_2^{[1]}$$